

Hadoop HDFS Commands

Command Name	Command Purpose	Example — Practice Flow
<code>start-dfs.sh</code>	It starts the main HDFS daemons: NameNode , DataNode , SecondaryNameNode	<code>start-dfs.sh</code>
<code>Jps</code>	Checks whether Hadoop services/daemons are running.	Check Hadoop services before starting HDFS practice: <code>jps</code>
<code>hdfs dfs -ls</code>	Lists files and folders in HDFS. Sub-purpose: List a specific directory. Sub-purpose: List recursively using <code>-R</code> .	List HDFS root: <code>hdfs dfs -ls /</code> List <code>myfolder1</code> : <code>hdfs dfs -ls /myfolder1</code> Recursive listing: <code>hdfs dfs -ls -R /data</code>
<code>hdfs dfs -mkdir</code>	Creates one or more directories/folders in HDFS.	Create <code>myfolder1</code> : <code>hdfs dfs -mkdir /myfolder1</code> Create <code>myfolder2</code> and <code>myfolder3</code> together: <code>hdfs dfs -mkdir /myfolder2 /myfolder3</code> Create subfolders: <code>hdfs dfs -mkdir /myfolder1/subfold1 /myfolder1/subfold2</code>
<code>hdfs dfs -touchz</code>	Creates one or more empty files in HDFS.	Create <code>myfile</code> : <code>hdfs dfs -touchz /myfile</code> Create <code>file1</code> inside <code>myfolder2</code> : <code>hdfs dfs -touchz /myfolder2/file1</code> Create multiple files: <code>hdfs dfs -touchz /myfolder3/file2 /myfolder3/file3</code>

echo	Creates a local file containing one line of content. It works on the Ubuntu/Linux filesystem, not directly in HDFS.	Create a local file: <code>echo "Hello Hadoop" > file1.txt</code> Then import it to HDFS using <code>-put</code>.
cat	Reads/displays the contents of a local Linux file. It can also be used to create a local multi-line file using <code>cat ></code>.	Read local file: <code>cat file1.txt</code> Create multi-line file: <code>cat > file2.txt</code> Type multiple lines → press Ctrl+D.
hdfs dfs -put	Imports/copies files or folders from the local Ubuntu filesystem → HDFS. Sub-purpose: Import one file. Sub-purpose: Import multiple files. Sub-purpose: Import an entire folder. Sub-purpose: <code>-f</code> can force overwrite.	Import one file: <code>hdfs dfs -put file1.txt /myfolder2/</code> Multiple files: <code>hdfs dfs -put file1.txt file2.txt file3.txt /myfolder2/</code> Folder: <code>hdfs dfs -put mydata /myfolder2/</code> Overwrite: <code>hdfs dfs -put -f sample.txt /data/input/</code>
hdfs dfs -cat	Reads/displays the contents of a file stored in HDFS.	Read <code>file1.txt</code> from HDFS: <code>hdfs dfs -cat /myfolder2/file1.txt</code> Read multi-line file: <code>hdfs dfs -cat /myfolder3/file2.txt</code>
hdfs dfs -cp	Copies a file or folder within HDFS. The original remains in the source location.	Copy <code>file1</code> from <code>myfolder2</code> to <code>subfold1</code>: <code>hdfs dfs -cp /myfolder2/file1 /myfolder1/subfold1/</code> Original <code>file1</code> remains in <code>/myfolder2</code>.
hdfs dfs -mv	Has two purposes: 1. Move a file/folder within HDFS. 2. Rename a file/folder within HDFS.	Move: <code>hdfs dfs -mv /myfolder3/file2 /myfolder1/subfold2/</code> Rename: <code>hdfs dfs -mv /myfolder3/file3 /myfolder3/file_three</code>
hdfs dfs -rm	Removes/deletes files from HDFS. Can remove one or multiple files.	Remove one file: <code>hdfs dfs -rm /myfile</code> Remove multiple files: <code>hdfs dfs -rm /file1 /file2 /file3</code>

<code>hdfs dfs -rmdir</code>	Removes an empty folder/directory. It will not remove a folder containing files or subfolders.	After removing all files from <code>subfold1</code> and <code>subfold2</code>: <pre>hdfs dfs -rmdir /myfolder1/subfold1</pre> Then remove <code>myfolder1</code> after it becomes empty.
<code>hdfs dfs -rm -r</code>	Removes a folder and everything inside it recursively. Used when the folder contains files/subfolders.	Example structure: <pre>/myfolder3/file1 /myfolder3/file2 /myfolder3/subfolder/</pre> Remove everything together: <pre>hdfs dfs -rm -r /myfolder3</pre>

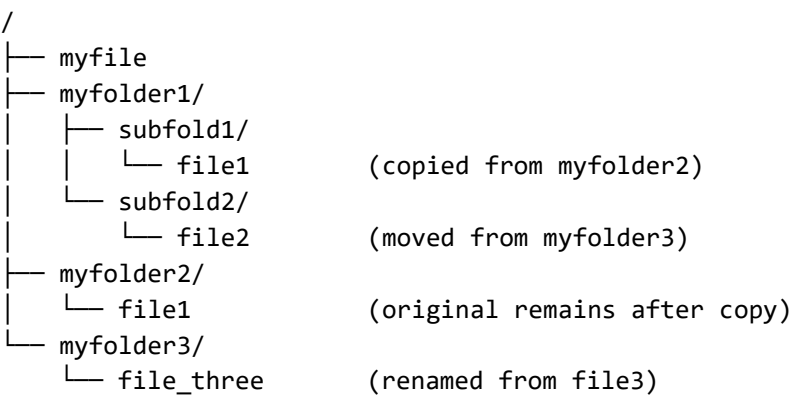
Hadoop NameNode Web UI

Use the NameNode Web UI to visually browse HDFS files and folders. **Open in a Windows web browser:**

`http://localhost:9870`

In the NameNode interface, use Utilities → Browse the file system to view the HDFS hierarchy.

HDFS Hierarchy



Running Hadoop in GitHub Codespaces

Step 1: Check running Docker containers : `docker ps`

Step 2: Start the Hadoop containers : `docker compose up -d`

Step 3: Run Hadoop commands inside the NameNode container

In Codespaces, Hadoop commands should be prefixed with: `docker exec namenode`

Example:

```
docker exec namenode hdfs dfs -ls /
```